

The empirical recurrence for OEIS A278278, recovered from its tail

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Abstract

OEIS A278278 records “Empirical recurrence of order 98” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 141$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 102$. Its last coefficient is $c_{98} = 455493888 \neq 0$, so no further tail satisfies anything shorter; any order-98 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A278278 is “Number of $n \times 6$ 0..1 arrays with every element both equal and not equal to some elements at offset $(-1,-1)$ $(-1,0)$ $(-1,1)$ $(0,-1)$ $(0,1)$ or $(1,0)$, with upper left element zero.”. It has offset 1 and begins

0, 109, 3723, 172091, 7466627, 329985827, 14526103064, 640011857446, ...

The entry states:

Empirical recurrence of order 98 (see link above)

As of the “Last modified” line on the live entry (Nov 16 2016 12:08:59, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 141$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 141$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 98: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 98.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 282$ exact terms from $n = 4$ on, it returns order 98 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{98} c_i a(n-i),$$

c_1	36	c_{26}	−161071123849182705	c_{51}	−88174042223680702509	c_{76}	3209499264
c_2	528	c_{27}	−7197239532238301	c_{52}	3614472005876887085674	c_{77}	−2447254857
c_3	−6347	c_{28}	578540247069874155	c_{53}	−3037266048257484857343	c_{78}	10650767408
c_4	−68394	c_{29}	−2420264558877085389	c_{54}	−26461955689442254023	c_{79}	−233179434
c_5	434782	c_{30}	5217821986871333417	c_{55}	2104063785509436184260	c_{80}	−37910705
c_6	3868992	c_{31}	−5539583469602835545	c_{56}	−2931793430096096267992	c_{81}	511792095
c_7	−16543278	c_{32}	836059708732631758	c_{57}	−23043951580802181897	c_{82}	−428649928
c_8	−92002400	c_{33}	18184110749459979067	c_{58}	3427063120648421695833	c_{83}	132575442
c_9	380363476	c_{34}	−46838800433477597284	c_{59}	−3134179816059787841900	c_{84}	−33436958
c_{10}	17271240	c_{35}	51866785861580827278	c_{60}	880994506258257129292	c_{85}	−2589022
c_{11}	−3467436397	c_{36}	−45696065731021838713	c_{61}	1885087206680018807597	c_{86}	186325364
c_{12}	40812995780	c_{37}	−5716691094004726757	c_{62}	−3972380424003215282921	c_{87}	−70464213
c_{13}	−109799432916	c_{38}	120571263640397406460	c_{63}	3409104965871551960374	c_{88}	8937365
c_{14}	−434707071211	c_{39}	−134333367735602047159	c_{64}	−1249217402090822157830	c_{89}	22333931
c_{15}	3069250395770	c_{40}	214769944137090408063	c_{65}	−944202240303141415077	c_{90}	4304914
c_{16}	−6252521473991	c_{41}	−285713545380445405587	c_{66}	2181027297728892990100	c_{91}	−2450538
c_{17}	−5355575762190	c_{42}	−10367452684779942327	c_{67}	−1953984455952266410693	c_{92}	−324518
c_{18}	74720773706885	c_{43}	139678041888306318896	c_{68}	1033128179992428113185	c_{93}	−143908
c_{19}	−349021173205035	c_{44}	−215711673043094078780	c_{69}	−213927091942262374044	c_{94}	1811241
c_{20}	700235676469641	c_{45}	238429498673645302989	c_{70}	−281352918708053933017	c_{95}	−554044
c_{21}	−289977451908487	c_{46}	726427671049092320215	c_{71}	446551916789177751783	c_{96}	709184
c_{22}	−2769077361076201	c_{47}	−1021337050269824973400	c_{72}	−352012461659383964476	c_{97}	−55402
c_{23}	17223891637510176	c_{48}	−343962514201332248690	c_{73}	168799692523860983035	c_{98}	45549
c_{24}	−46445878258058931	c_{49}	1033431261218133528760	c_{74}	−42437504378919108656		
c_{25}	103919897210511825	c_{50}	−1305039345328678420518	c_{75}	−15623252594429297254		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 102$, and it is the recurrence OEIS A278278 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{98} - c_1x^{98-1} - \dots - c_{98}$. Its constant term is $-c_{98} = -455493888$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 98 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 98, so $\deg q = 98$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 98, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 98, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 455493888.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-98 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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